

Mobile HCI

Lecture 2: Design principles and interaction techniques

Carla F. Griggio
Tenure-track Assistant Professor
cfg@cs.aau.dk

Agenda

- Analysis of Interaction Techniques
- Designing for Context
- Field studies

Interaction Qualities

What aspect (quality) of interaction do we want to improve with an interaction technique?

Examples:

Discoverability: How to help users discover available but hidden interactions and commands? (e.g., gestures)

Input Expressiveness: How to increase the range of available input possibilities?

Performance: How to help users perform tasks faster, or avoid errors?

and more!

Interaction Techniques

Technology-dependent methods for performing (often generic) interactions tasks (e.g., typing, pointing, selecting). Each technique relies on one or more sensing technologies to combine input and output (feedback) modalities to perform an interaction task.

What’s a “**modality**”? A structured communication channel (between a human and an interactive technology) characterized by a specific human expressive or perceptual capability

Input Modalities	Sensing technologies	Interaction Tasks	Output Modalities
Single Touch (e.g., tap) Multi-touch (e.g., pinch) Touch gestures (e.g., swipe) Pressure/Force touch Voice/Audio Motion (e.g., shake) Pen/Stylus Text Mid-air/body gestures Lip reading Sign language Eye gaze Object recognition ...	Camera (front, back, 360, etc) Capacitive touch Pressure/Force sensing Microphone Accelerometer/Gyroscope Physical keyboards Touchscreen keyboards Infrared light sensors Radio frequency (e.g., NFC contactless payment) ...	Select Point Text entry Display commands Select/Execute command Scroll / Navigate screen Zoom ...	Visual Auditory Haptic ...

Hands-on!

Choose an interaction technique to present and discuss here: https://aaudk-my.sharepoint.com/:p:/g/personal/wb25qs_cs_aau_dk/IQBdHidOp-BOT4wWouw4eurqAVI5oea2RO1BvpU8m73f_U8?e=tTKK4u

Let's talk about *Context*

Designing for Context

According to ISO standard 13407 (1999), context of use is related to “user characteristics, task, as well as technical, physical, and social environment”.

Mobile technologies bring the opportunity and challenge of potentially being used under different contexts.

- **Opportunities?** We’re increasingly able to perform any kind of digital task regardless of where we are. We bring the technology with us, instead of going to where the technology is.
- **Challenges?** Designers can’t assume the context of use of a system, which can limit the effectiveness and overall usability of their designs.

Note: Mobile computing is especially concerned with context, but good design should *always* consider the *context of use* of a technology or system!

Types of Context: Physical

Physical context describes the apparent features of a situation in which the human-mobile computer interaction takes place, including spatial location, functional place and space, sensed environmental attributes, movements and mobility, the user's body, and other objects/artefacts present.

- location and altitude, physical objects (e.g., desk, doors), orientation (e.g., west) and weather and lighting conditions.
- the user's bodily capabilities and limitations, e.g., range of movement of fingers, vision accuracy, hearing impairments, etc.
- e.g., when developing context-aware applications, the information presented to a user is based on her/his geographical location.



You're going too fast!



Pokémon GO should not be played while driving.

I'M A PASSENGER



DRIVE SAFELY



Do not trespass while playing
Pokémon GO.

OK



NO TRESPASSING



Types of Context: Temporal

Temporal context describes the user's interaction with the mobile computer in relation to time in multiple ways such as duration, from time of day to years, the situation before and after use, actions in relation to time, and synchronism (e.g., real-time vs. asynchronous communication)

- factors of past and future situations, from time of day to the week, month or season
- time available for completing the task
- the actions in the temporal dimension vary and can be classified as hurried, normal or waiting

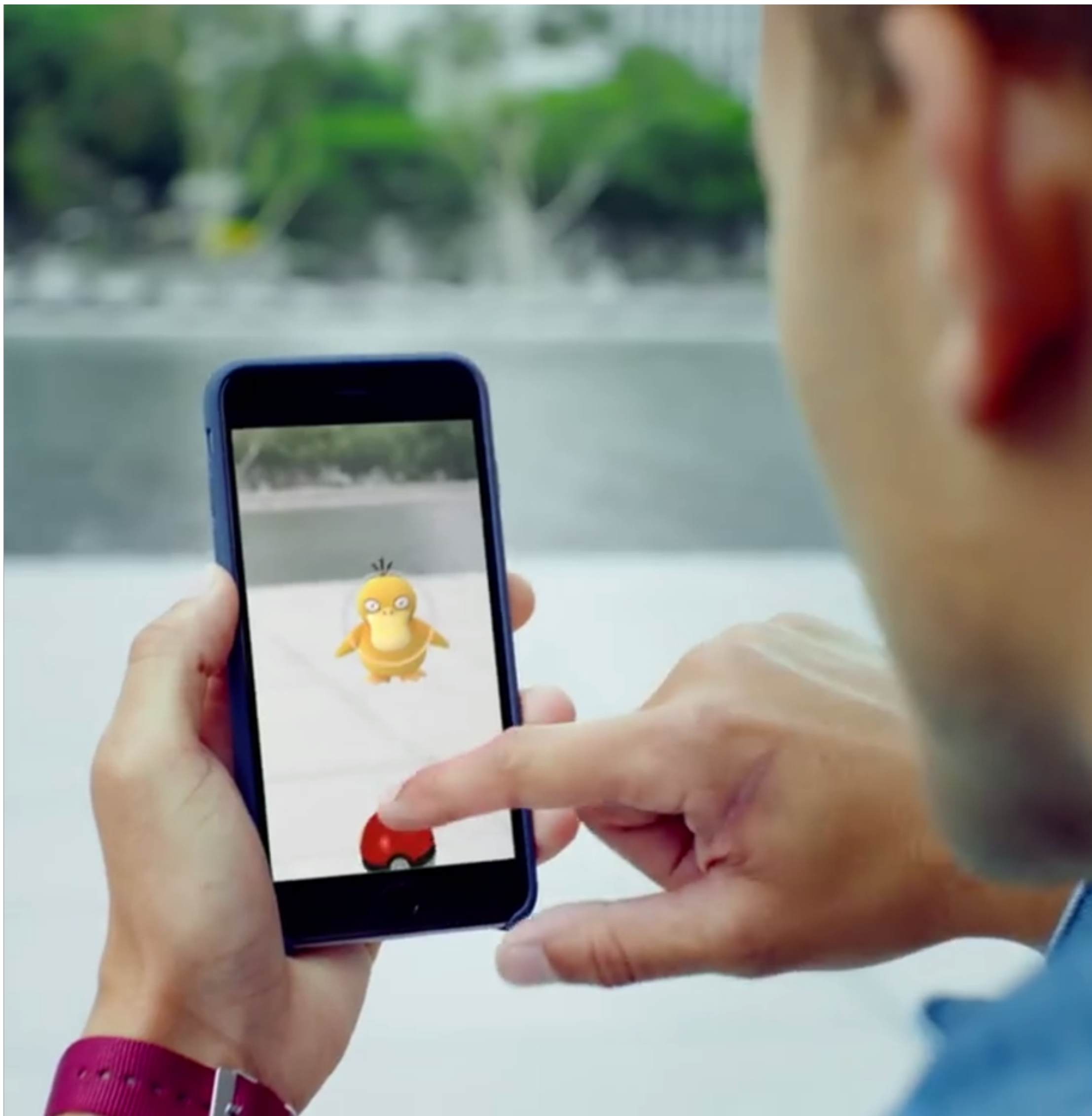
Pokémon GO



Types of Context: Task

Task context describes the surrounding tasks in relation to user's task of interacting with mobile computer containing the subcomponents of multitasking, interruptions and task domain. Task context is related to the demands of the entire situation upon one's attention.

In addition, studies of factors of task context can be identified which have divided them into **work or leisure related tasks** (see Agre, 2001); and according to the scope or **criticality** of a task, into **primary** and **secondary** tasks (Matthews et al., 2007); or as **foreground** and **background** interaction (Svanaes, 2001).



Foreground task



Background task

Types of Context: Social

Social context describes the other persons present, their characteristics and roles, the interpersonal interactions and the surrounding culture that influence the user's interaction with a mobile computer.

- other people's influence on the user and the user's social contribution goals (Roto, 2006), traditions (McCullough, 2001) privacy, relationship with the surrounding people, and the behaviour of the surrounding people (Dunlop & Bradley, 2005).





Do not trespass while playing
Pokémon GO.

OK



NO TRESPASSING



Types of Context: Technical

Technical context describes relation of other relevant systems and services including devices, applications and networks, their interoperability, informational artefacts or access, and mixed reality to the user's interaction with the mobile computer.

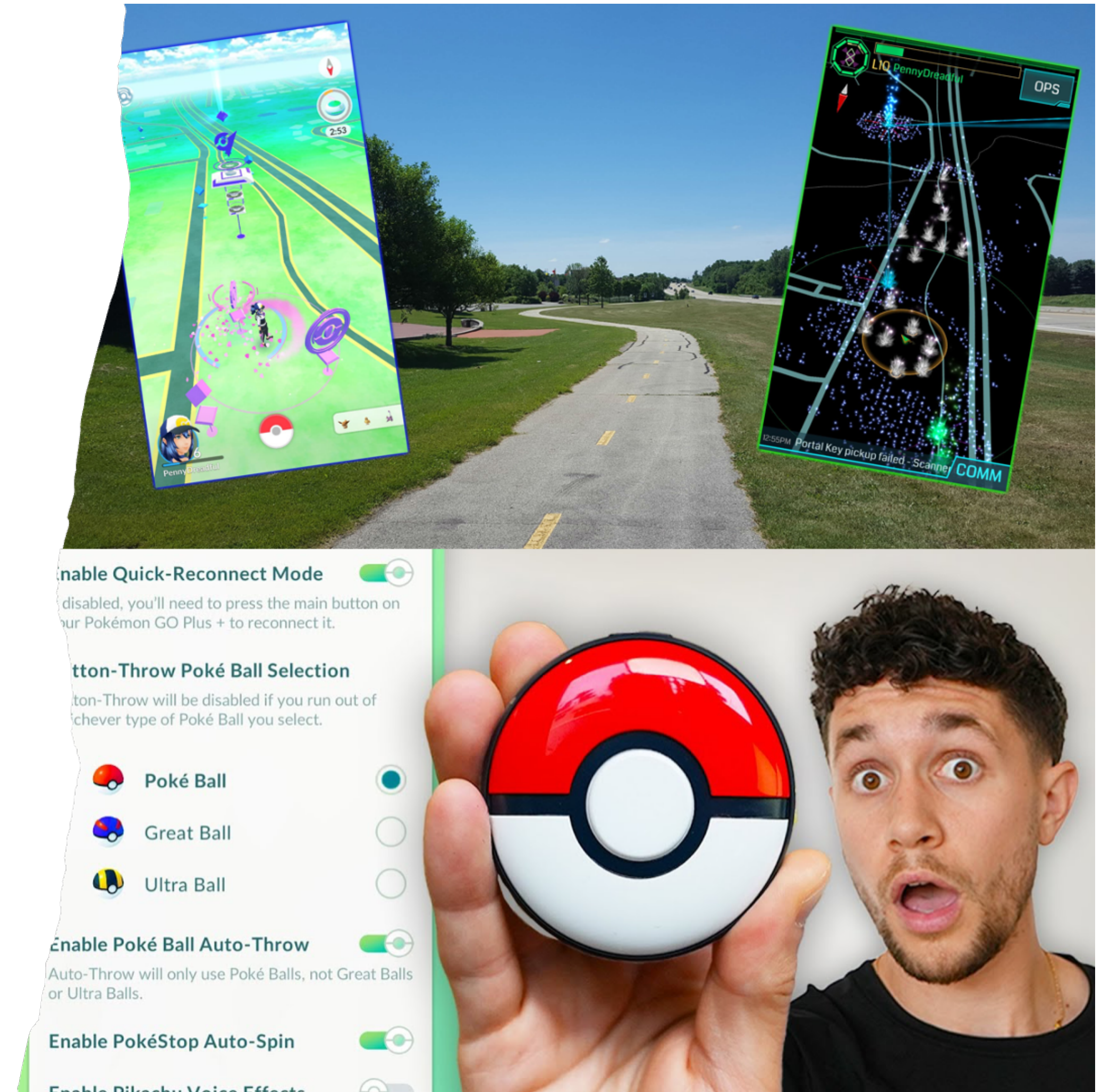
- devices, available infrastructure, facts and system assumptions (Petrelli et al., 2001), sensors (Fogarty et al., 2004) and network services (Winogard, 2001)

Types of Context: Technical

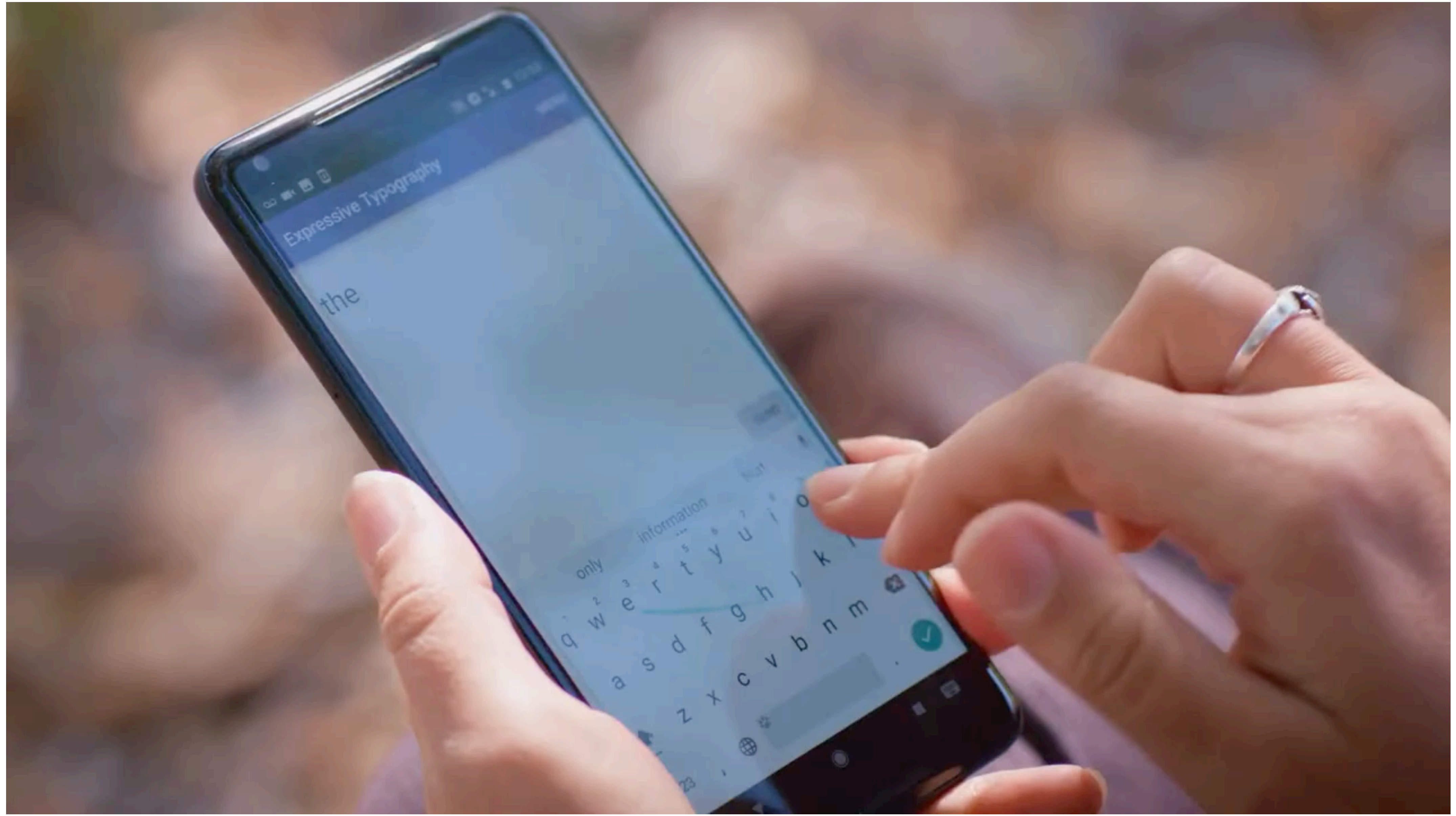
Based on OpenMapsStreet

Uses phone sensors (GPS, multitouch screen, step counter)

Extends with Pokemon Go Plus (Bluetooth)



What types of context may interfere with the usability of this technique?



What types of context may interfere with the usability of this technique?



Accessing functionality effeciently in one step

Augmenting Couples' Communication with Lifelines: Shared Timelines of Mixed Contextual Information

FIELD STUDY WITH 9 COUPLES

Carla F. Griggio, Midas Nouwens, Joanna McGrenere, and Wendy E. Mackay
CHI '19

Couples have special communication practices and needs



Express care for each other via many apps

(Cramer & Jacobs, CHI'15)



Need to stay in sync

(Thayer et al., CSCW '12)



Worry when the other cannot reply

(Bales et al., CSCW '11)

Apps rarely offer couple-specific functionality

**Design opportunity:
augment couples' communication with technology tailored to
their special practices and needs**

Sharing contextual information streams



Life360 Family Locator app



Location

Bales et al. 2011
Schildt et al. 2016

- increase feelings of connectedness
- find peace of mind



Motion

Bentley & Metcalf 2007

- coordinate everyday tasks
- infer availability
- start conversations

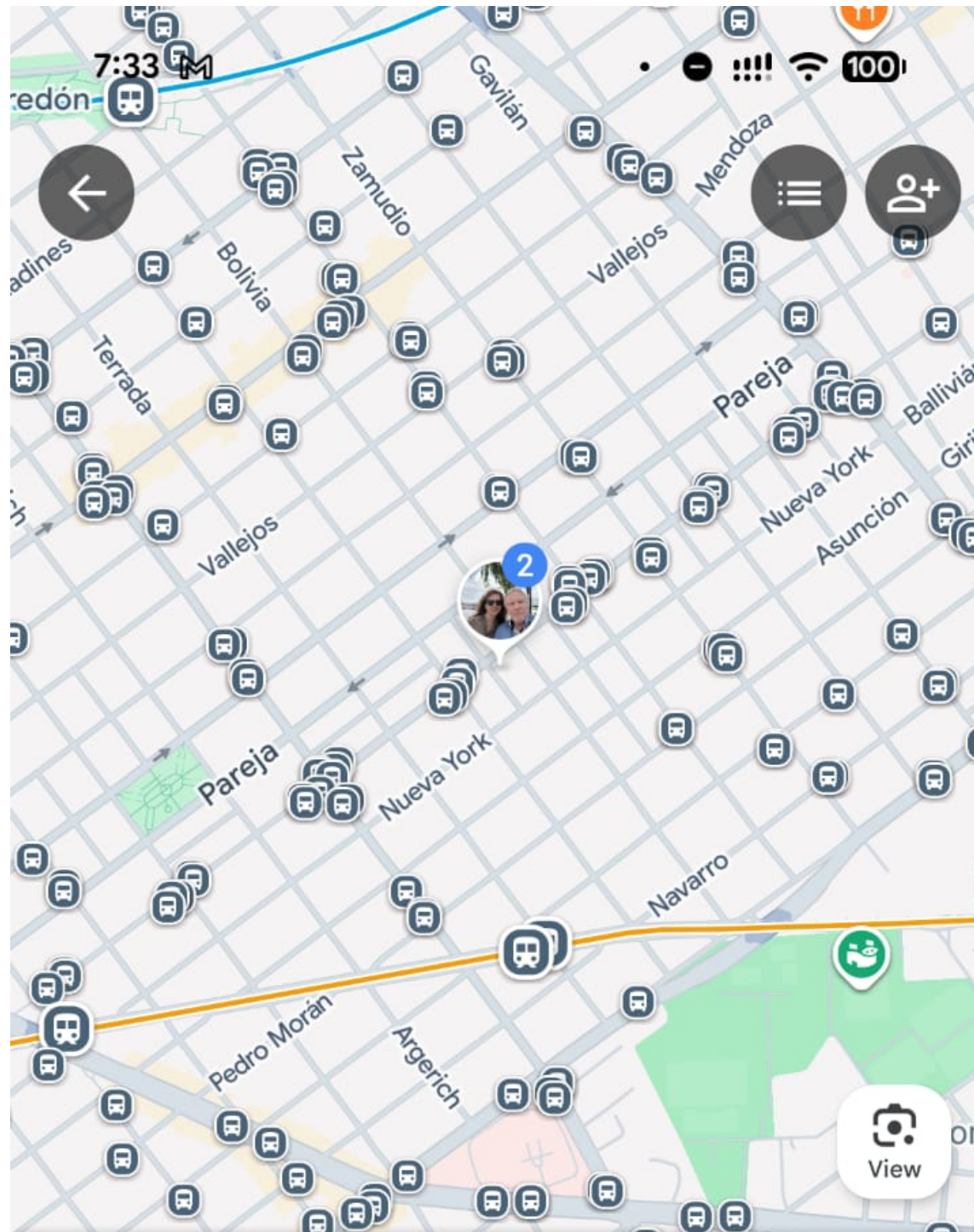


Heart rate

Hassib et al. 2017

- increase feelings of connectedness
- start conversations

Sharing contextual information streams



Silvana Andrea García Margalejo

12,069 km away · 60% · 2 min ago

Local time 3:33 AM

Tripoli 4097, C1419DSA Cdad.
Autónoma de Buenos Aires, Argentina

 Directions

Notifications

0 locations

 Add

Can see your location

Until you turn this off

 Stop



Location

Bales et al. 2011

Schildt et al. 2016

- increase feelings of connectedness
- find peace of mind



Motion

Bentley & Metcalf 2007

- coordinate everyday tasks
- infer availability
- start conversations



Heart rate

Hassib et al. 2017

- increase feelings of connectedness
- start conversations

Battery in Google Maps

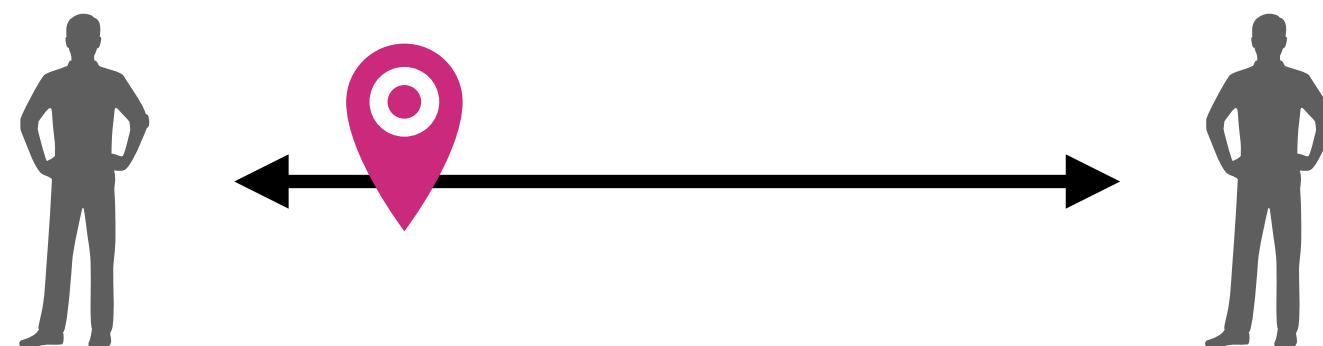
One stream at a time



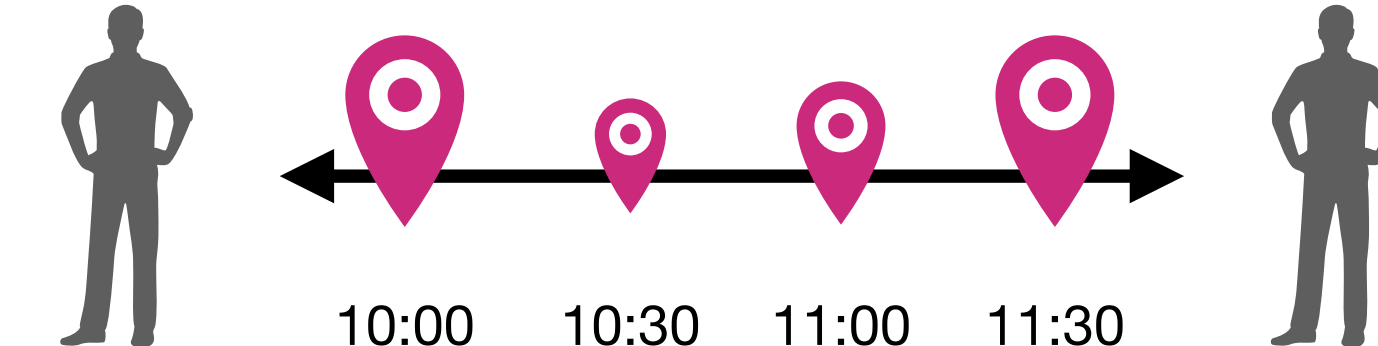
Mix of multiple streams



Live (ephemeral) data



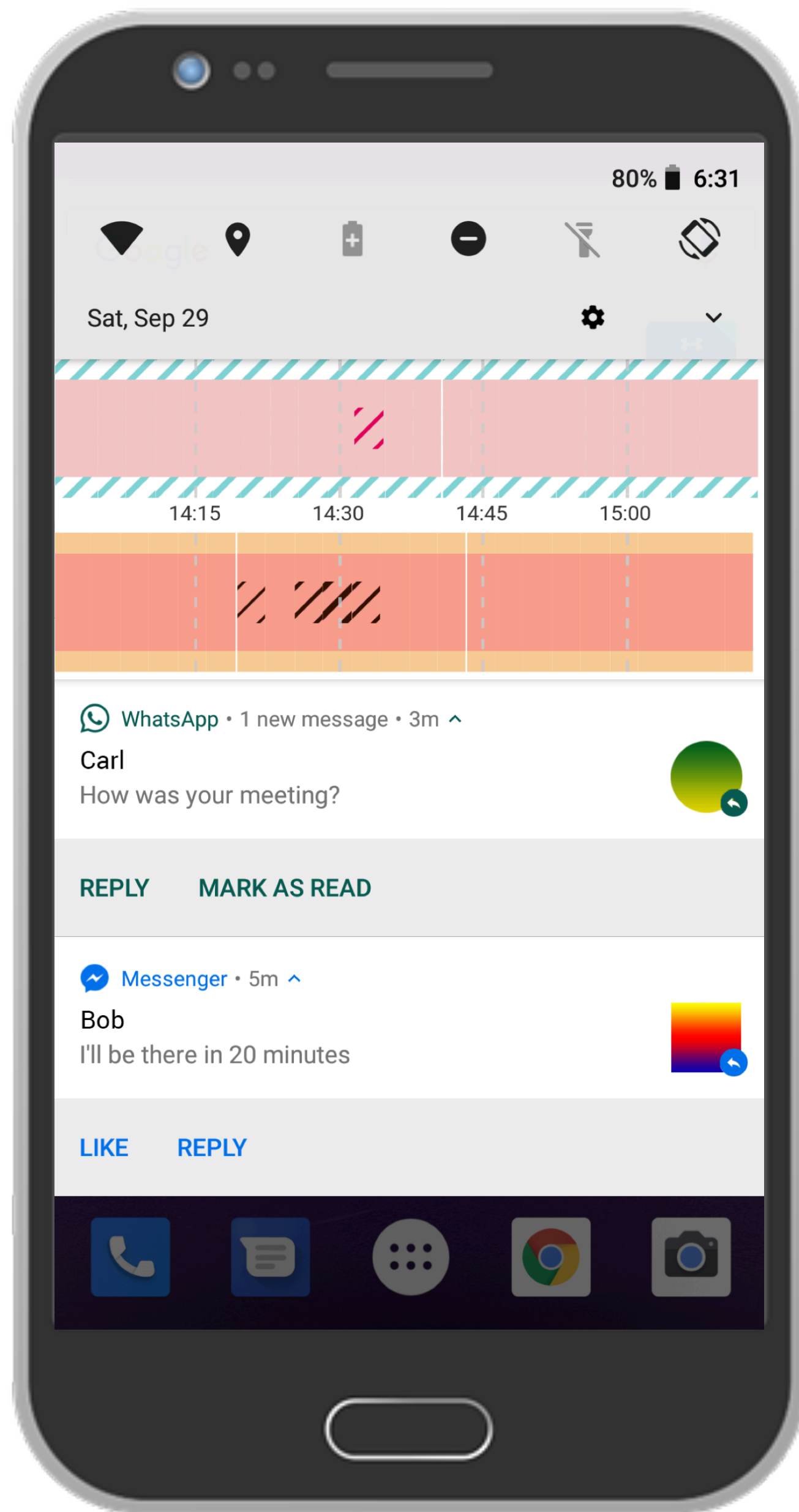
Persistent data



How does couples' communication change when sharing
multiple, persistent streams of contextual information?

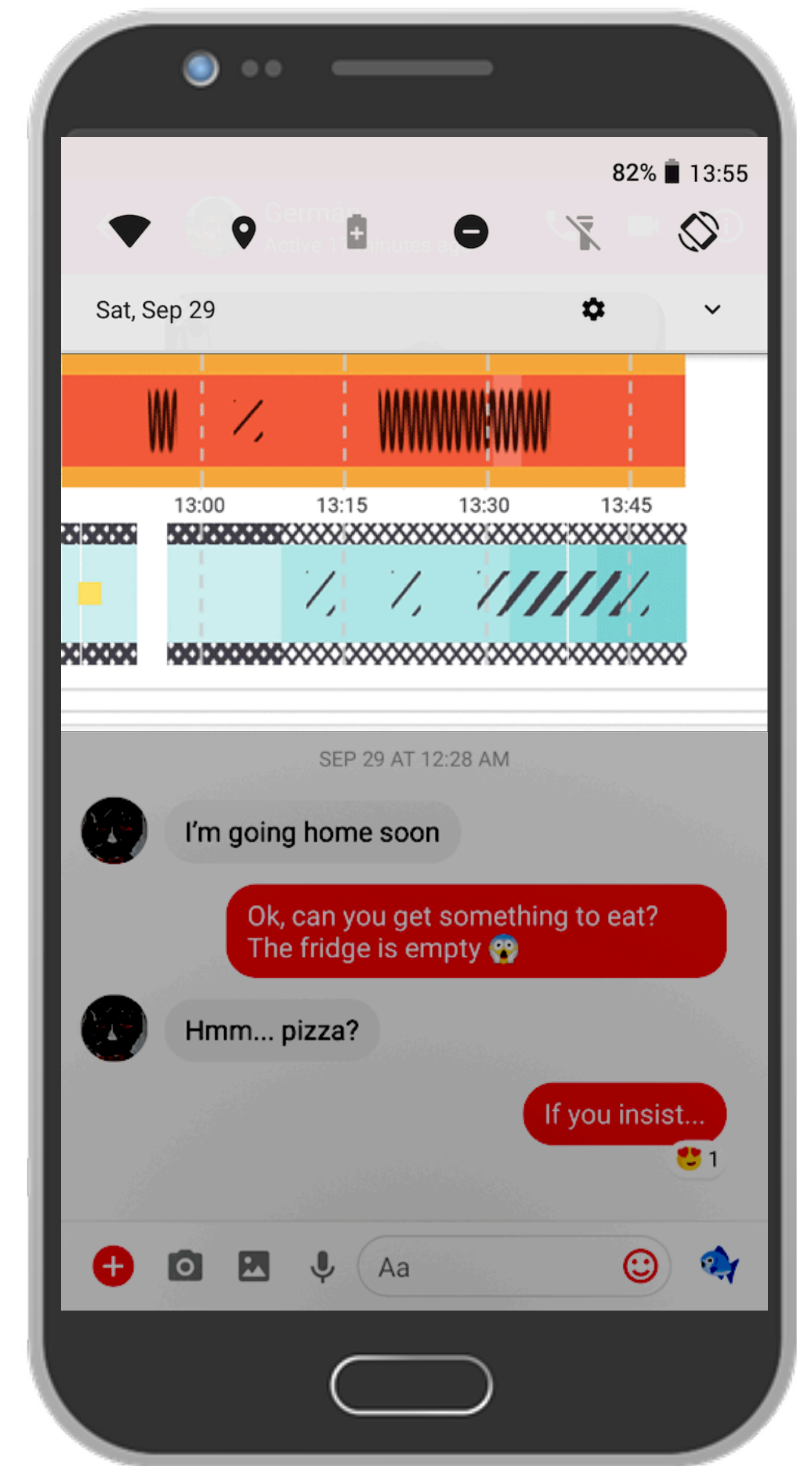
The Lifelines technology probe



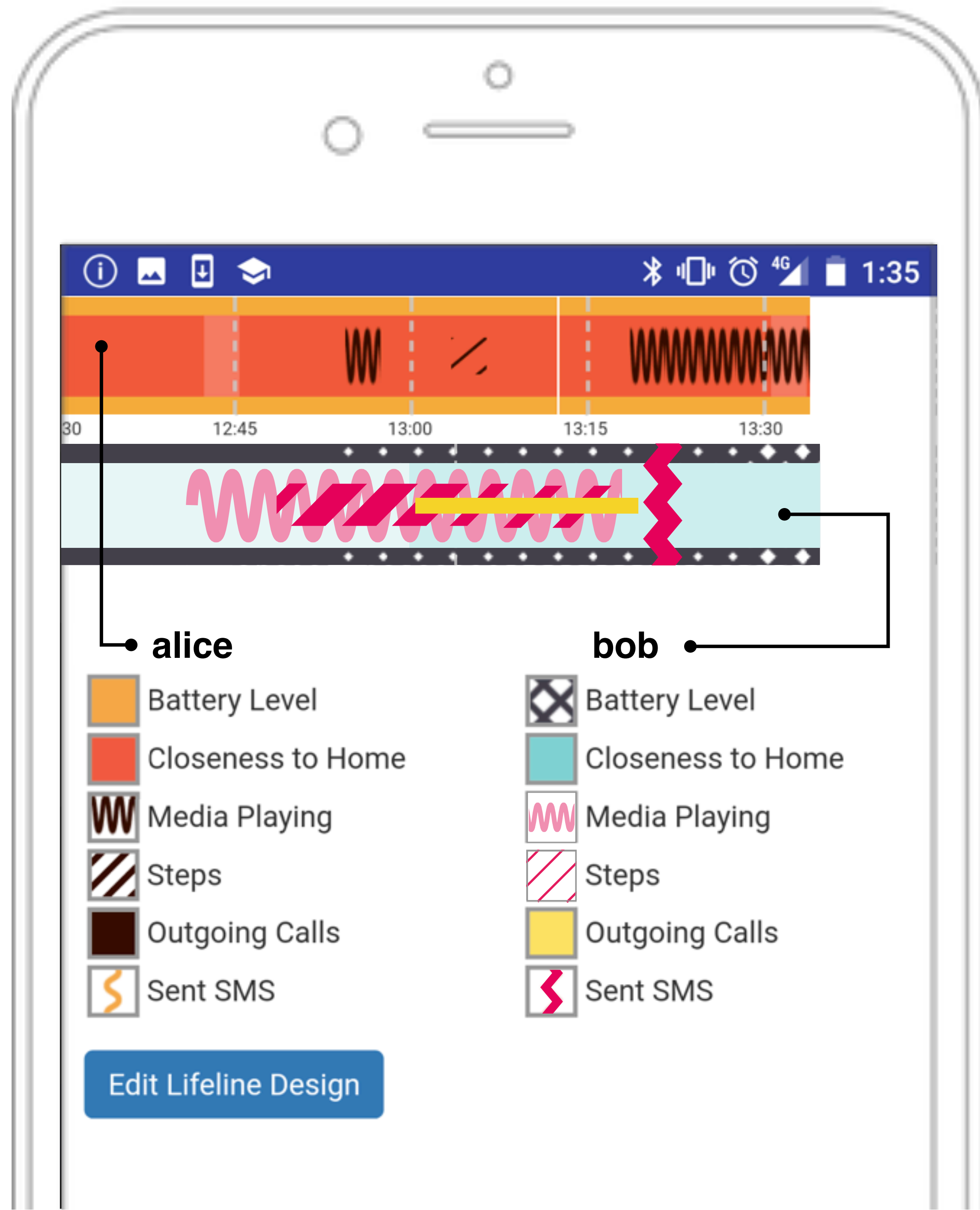


For peripheral awareness
in the notifications panel

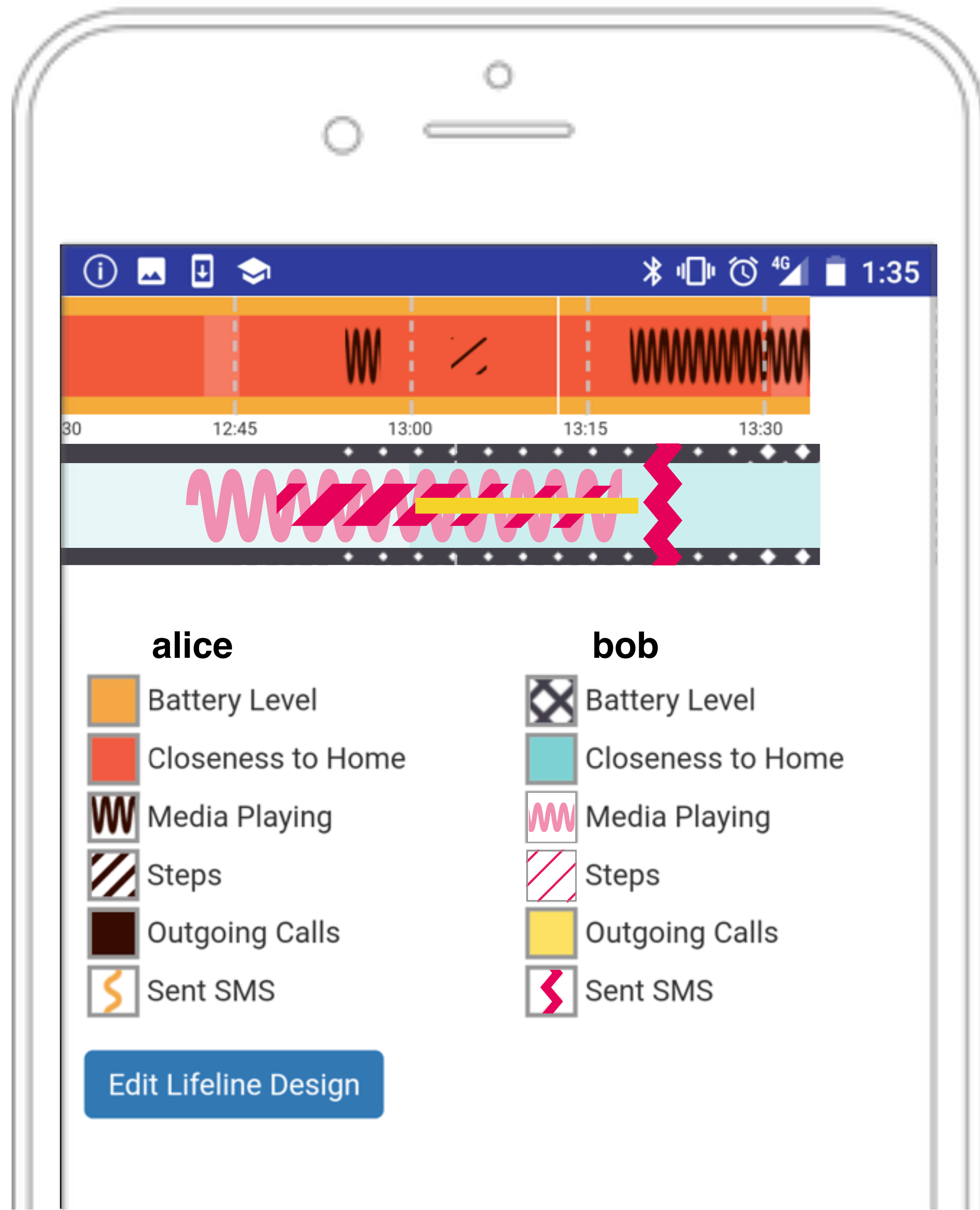
Lifelines Notification



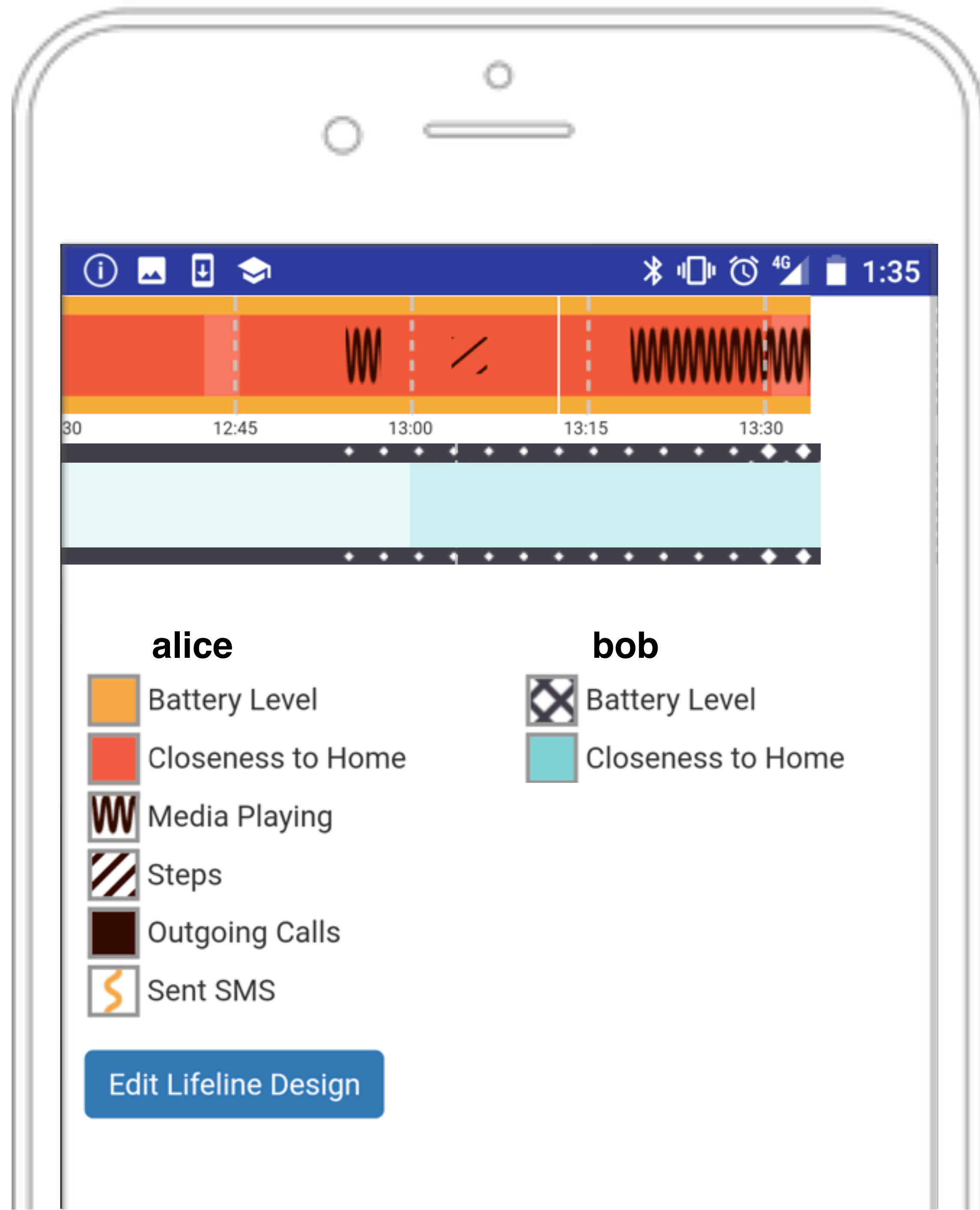
For augmenting conversations
with contextual information



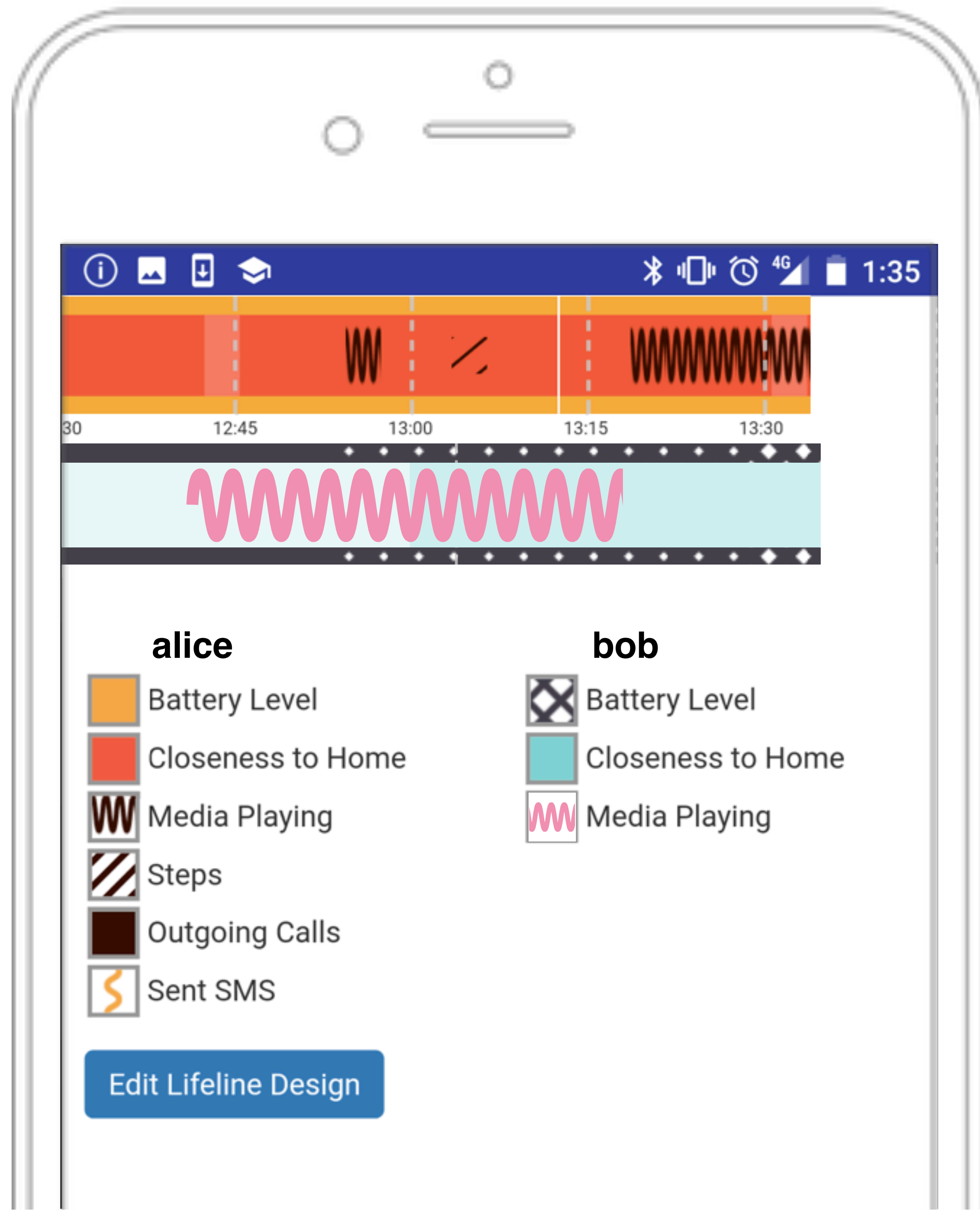
The Linebuilder



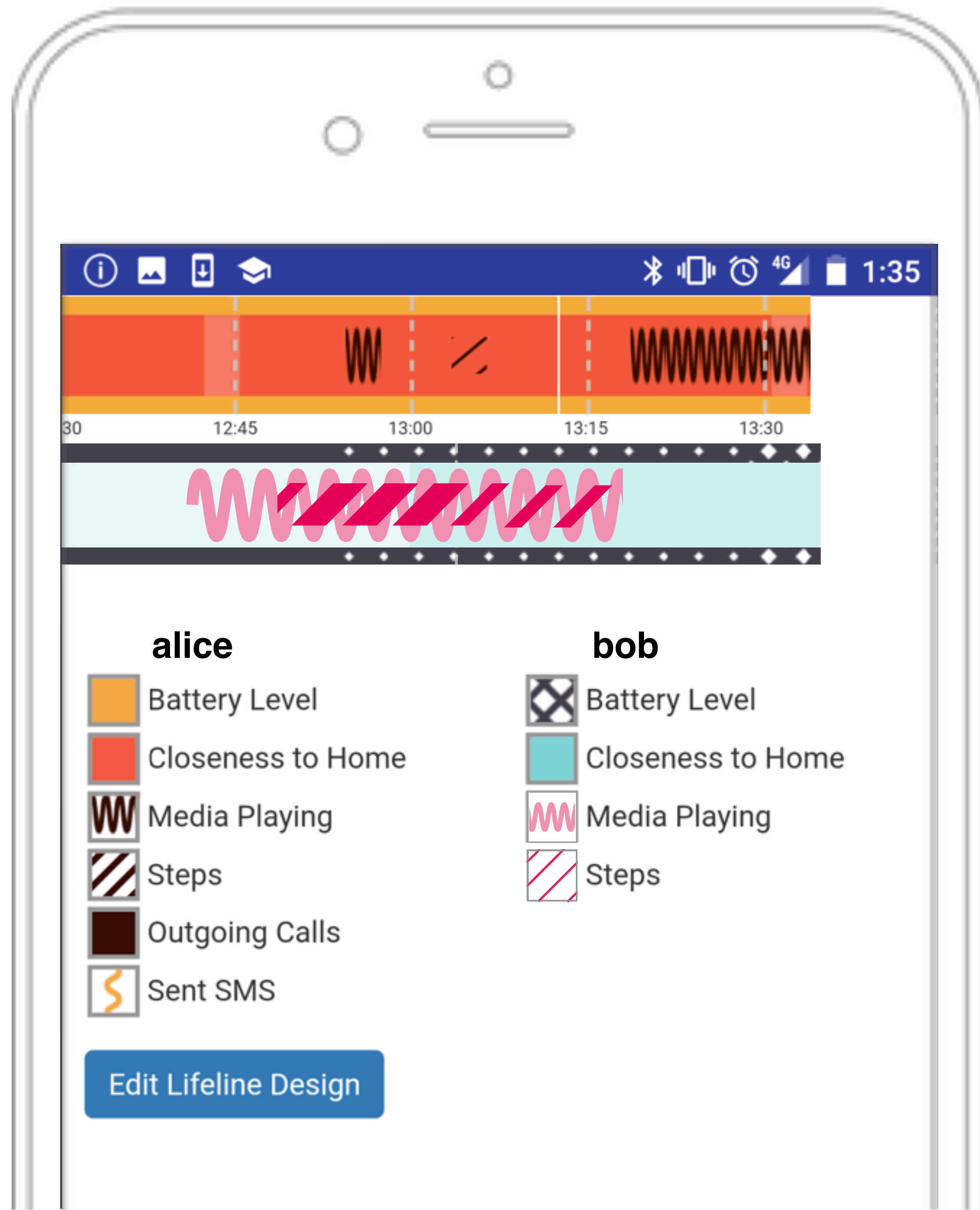
6 streams of contextual information



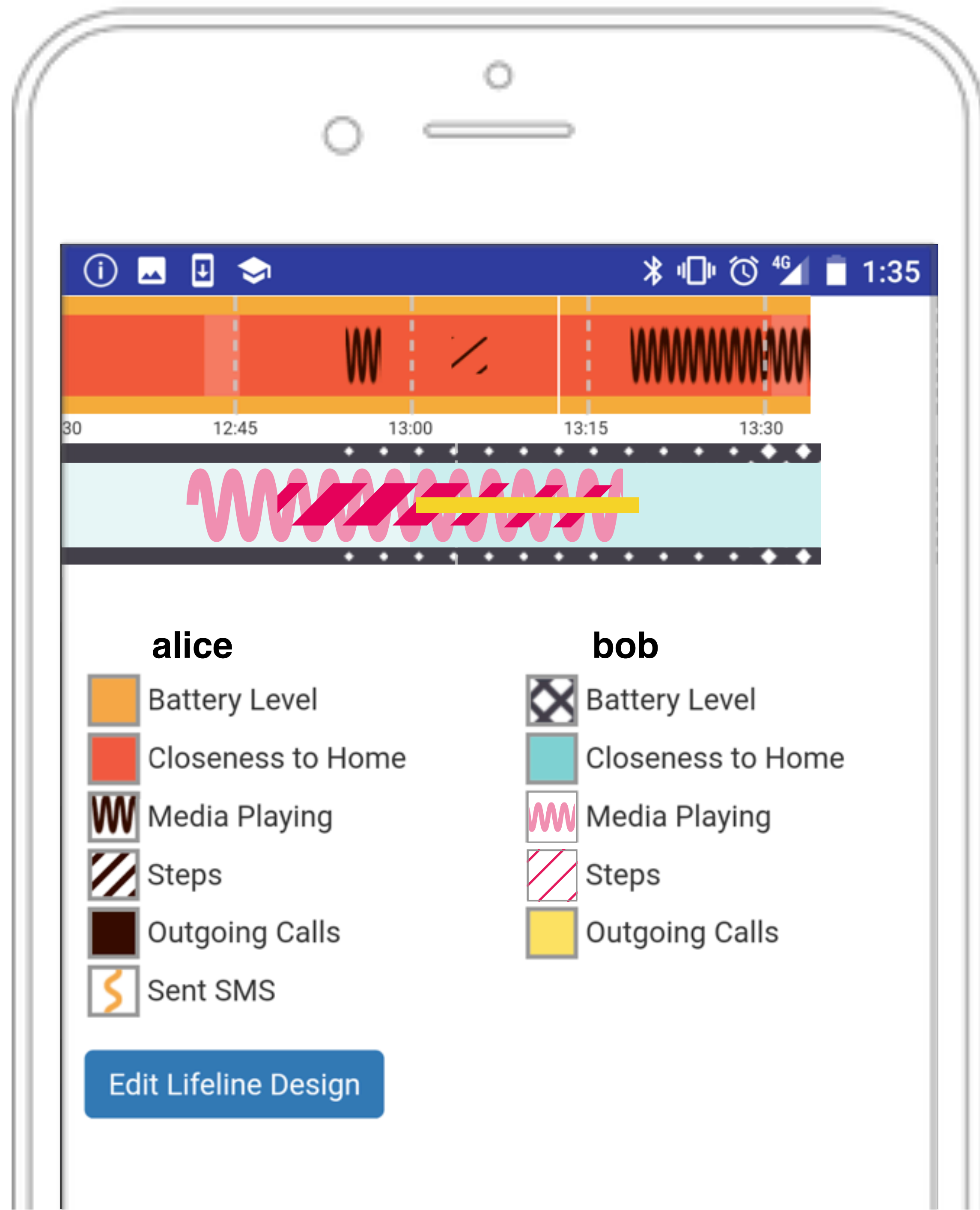
6 streams of
contextual
information



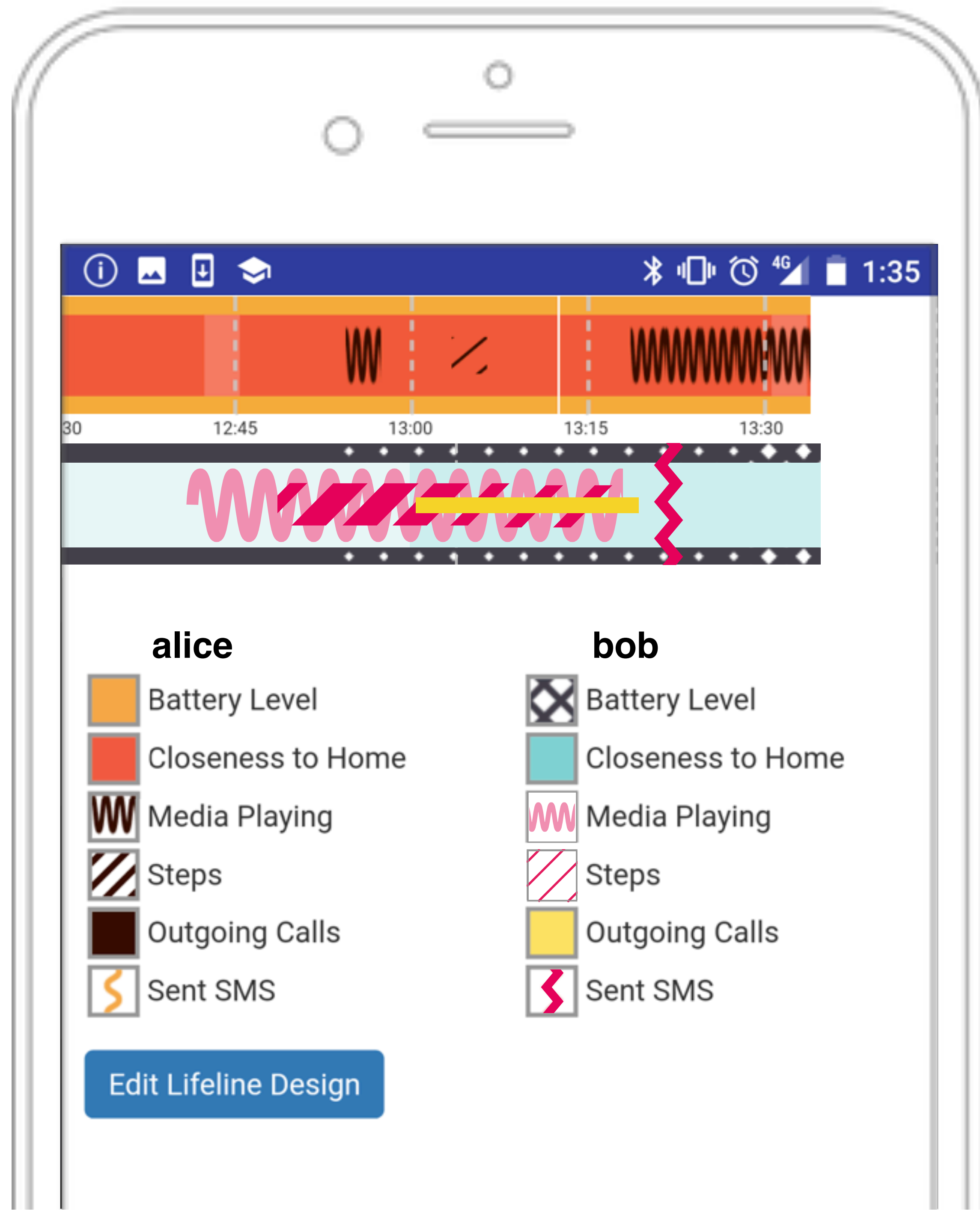
6 streams of
contextual
information



6 streams of
contextual
information



6 streams of contextual information



6 streams of
contextual
information



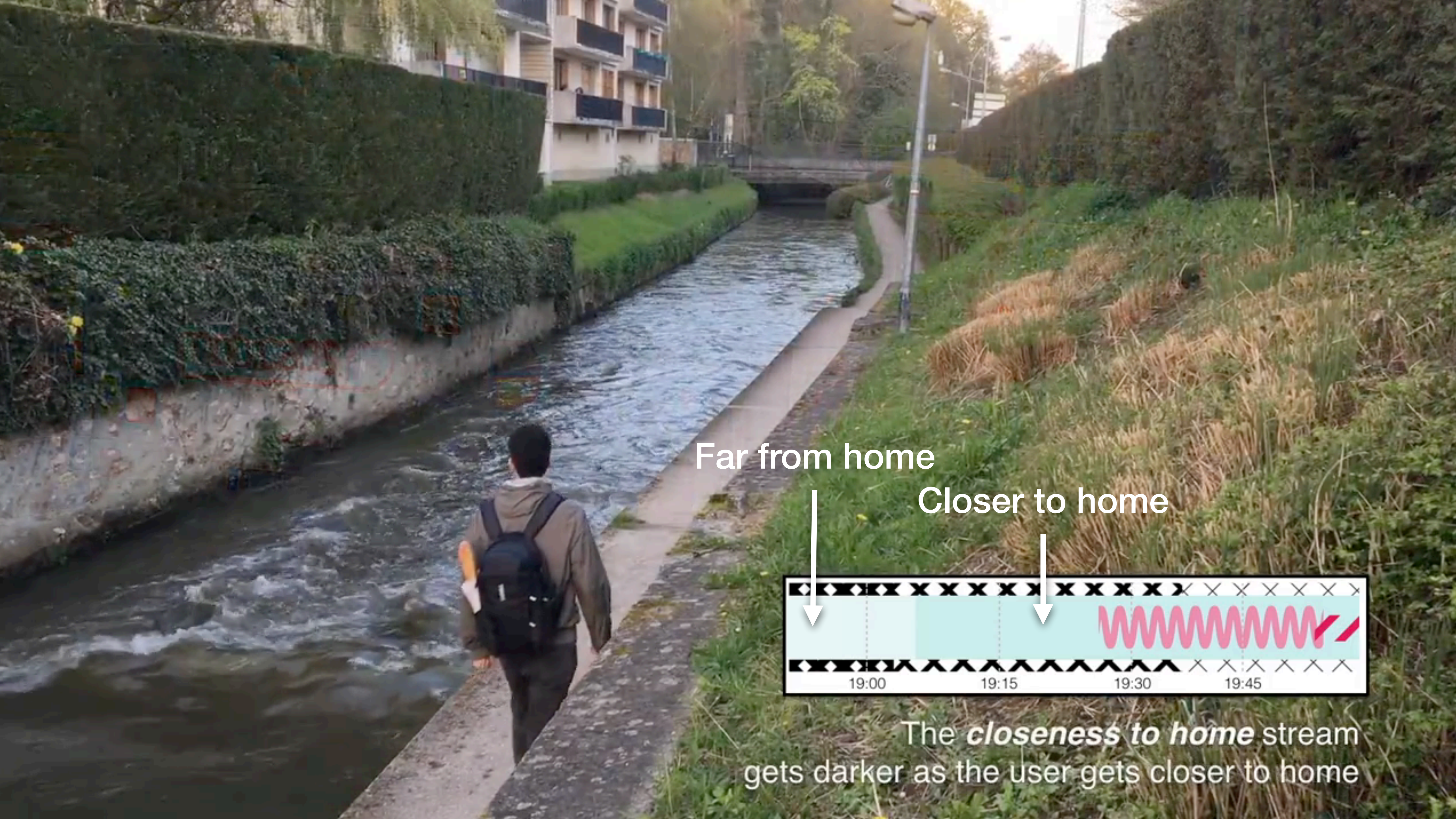
1 hour ago



now

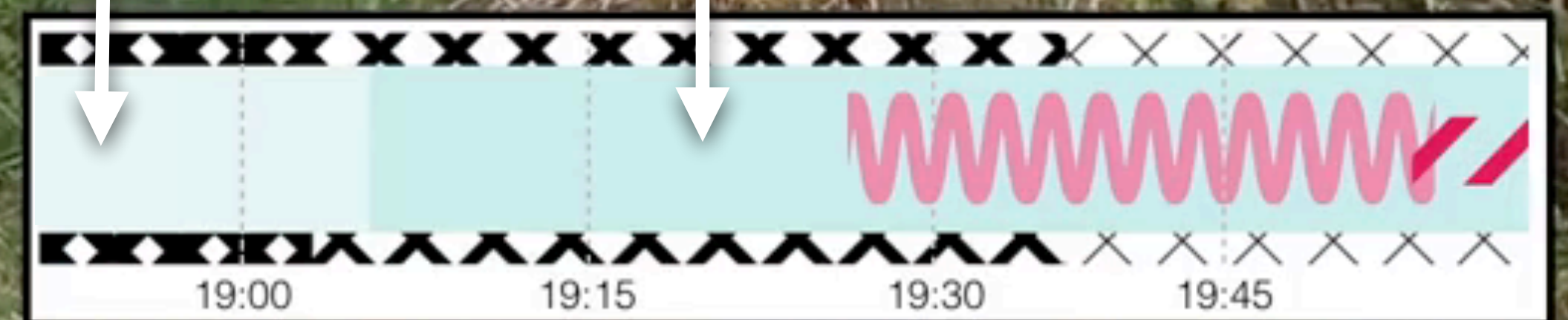


His ***battery level*** stream gets thinner as the phone's battery drops



Far from home

Closer to home



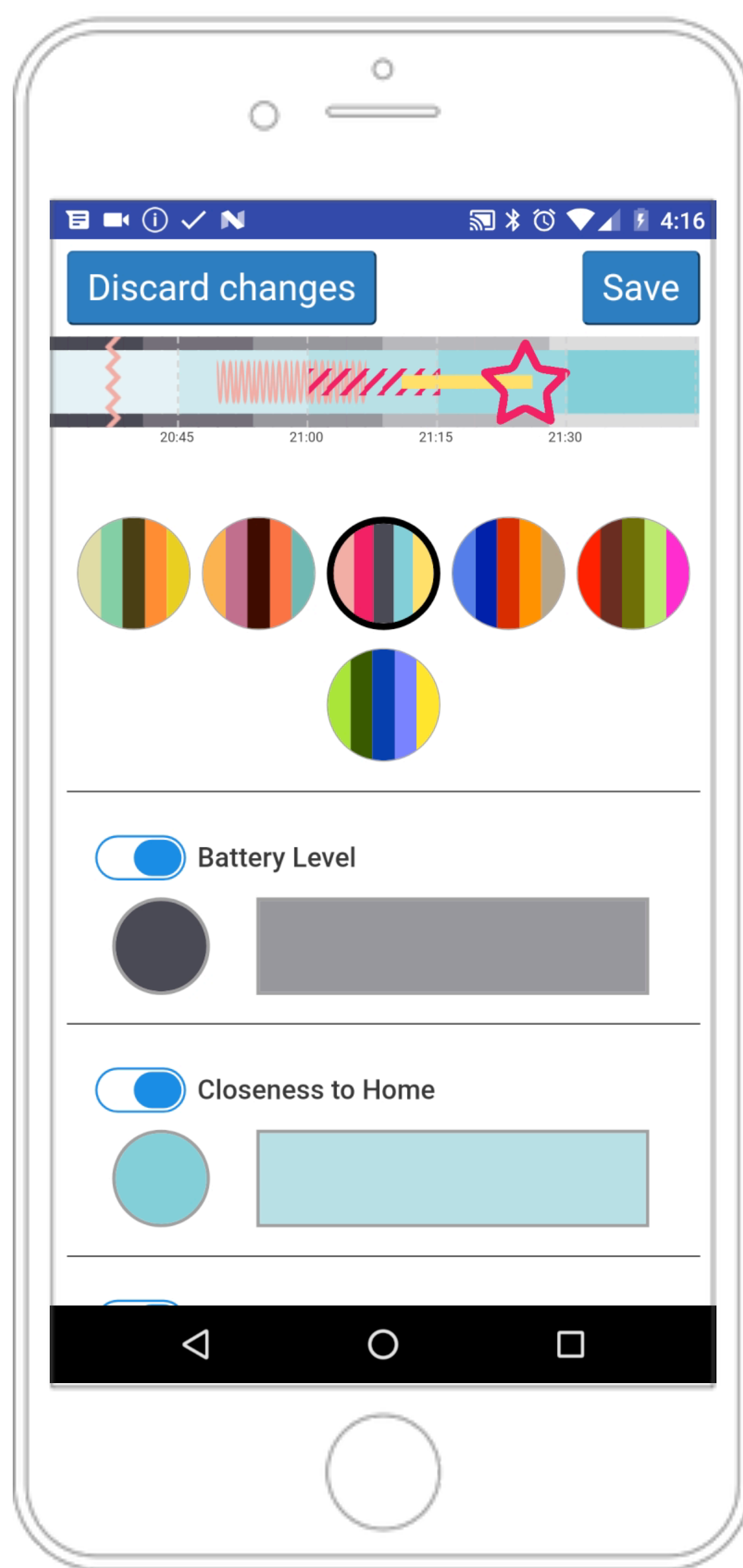
The *closeness to home* stream gets darker as the user gets closer to home



Very close
to home



The ***closeness to home*** stream
gets darker as the user gets closer to home



Partners individually choose
the streams they want to share with the other

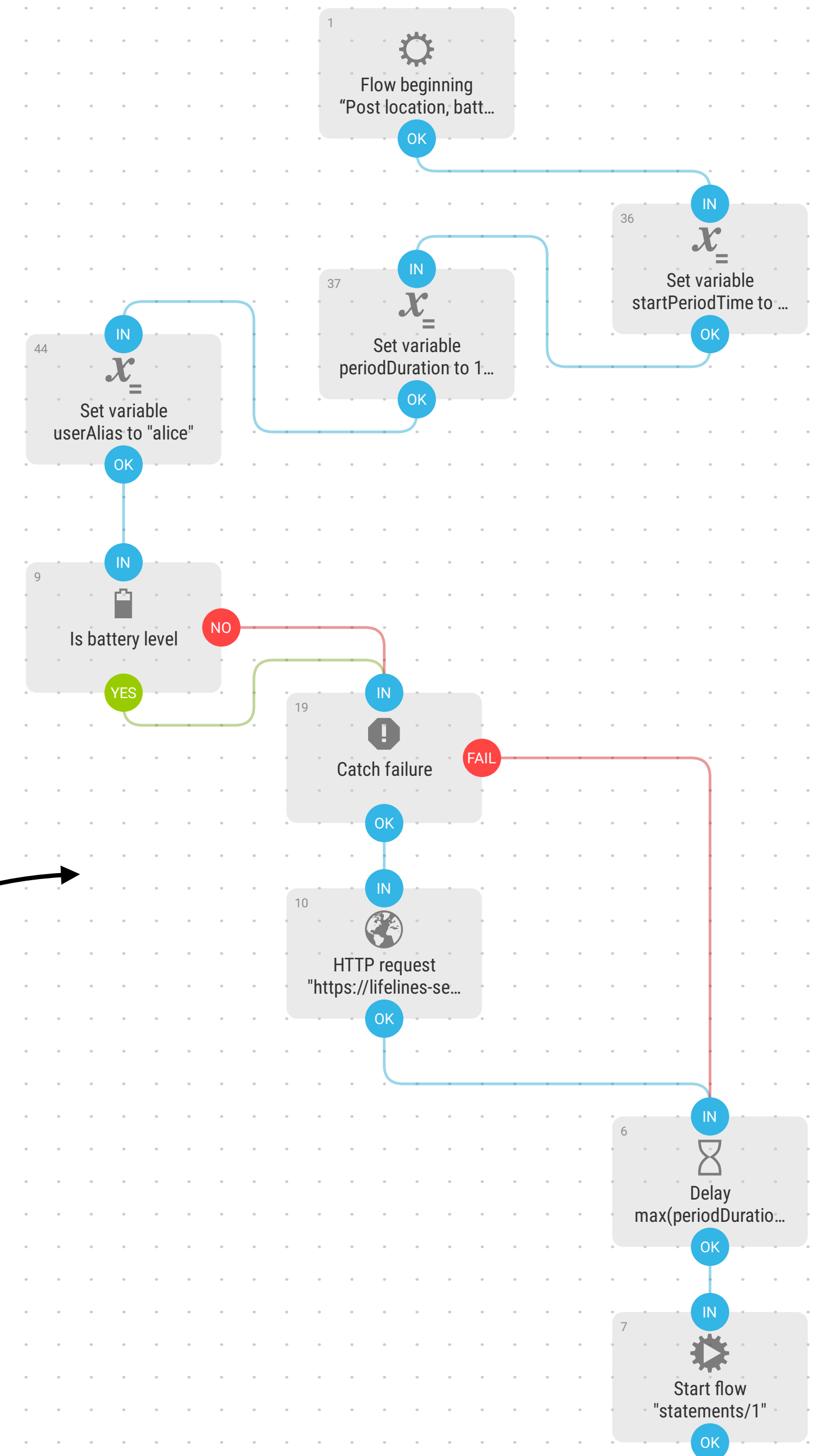
SET UP

Participants installed the Automate app for sensing and sharing their contextual data



Automate
Android App

Sensing and sharing
BATTERY LEVEL
every three minutes



METHOD

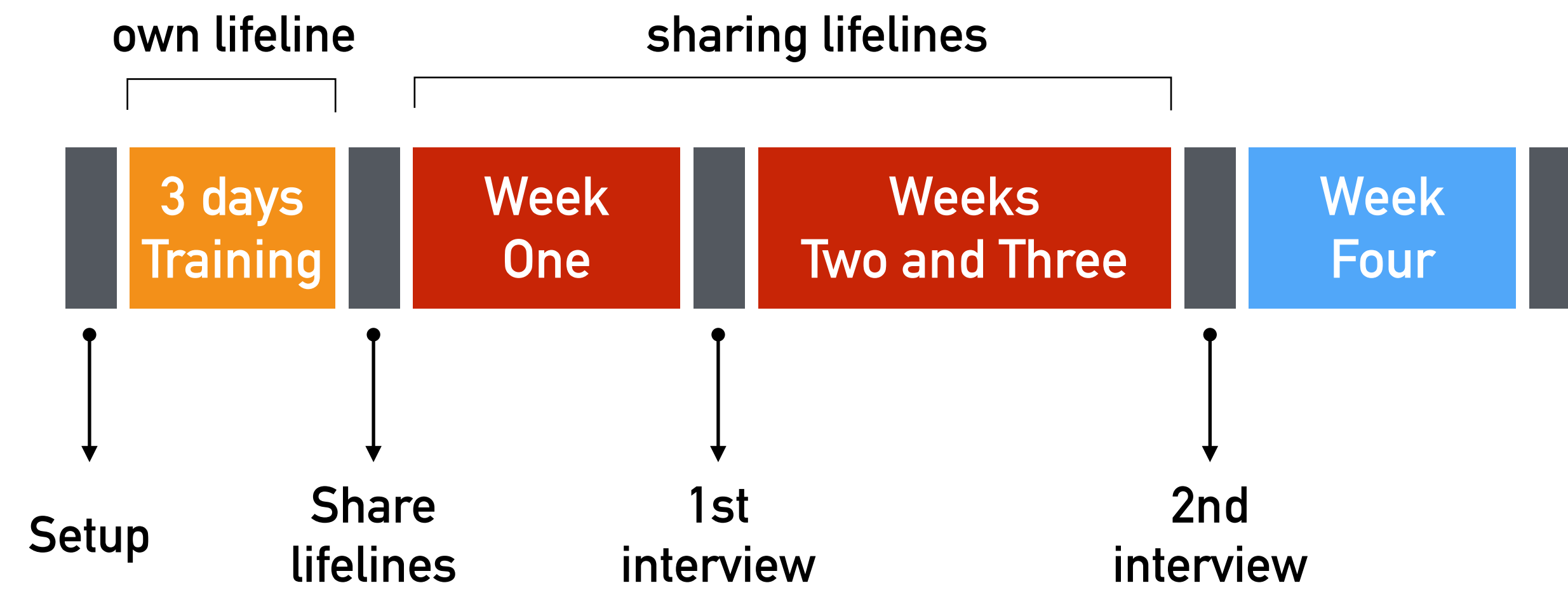
One-month technology-probe study

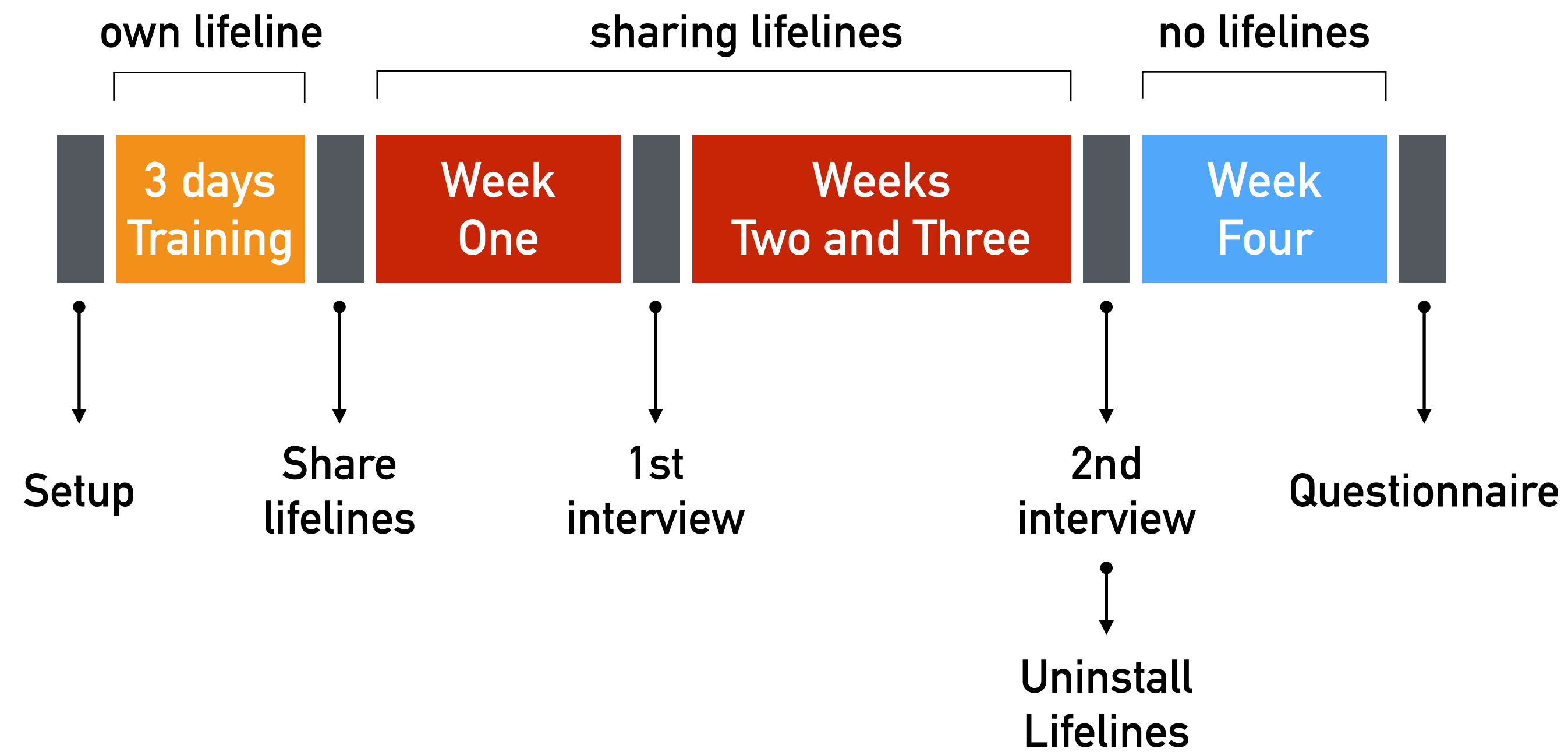
9 couples (8 living together)

Argentina, Brazil, UK, Switzerland, New Zealand & France

Setup and Interviews via Skype

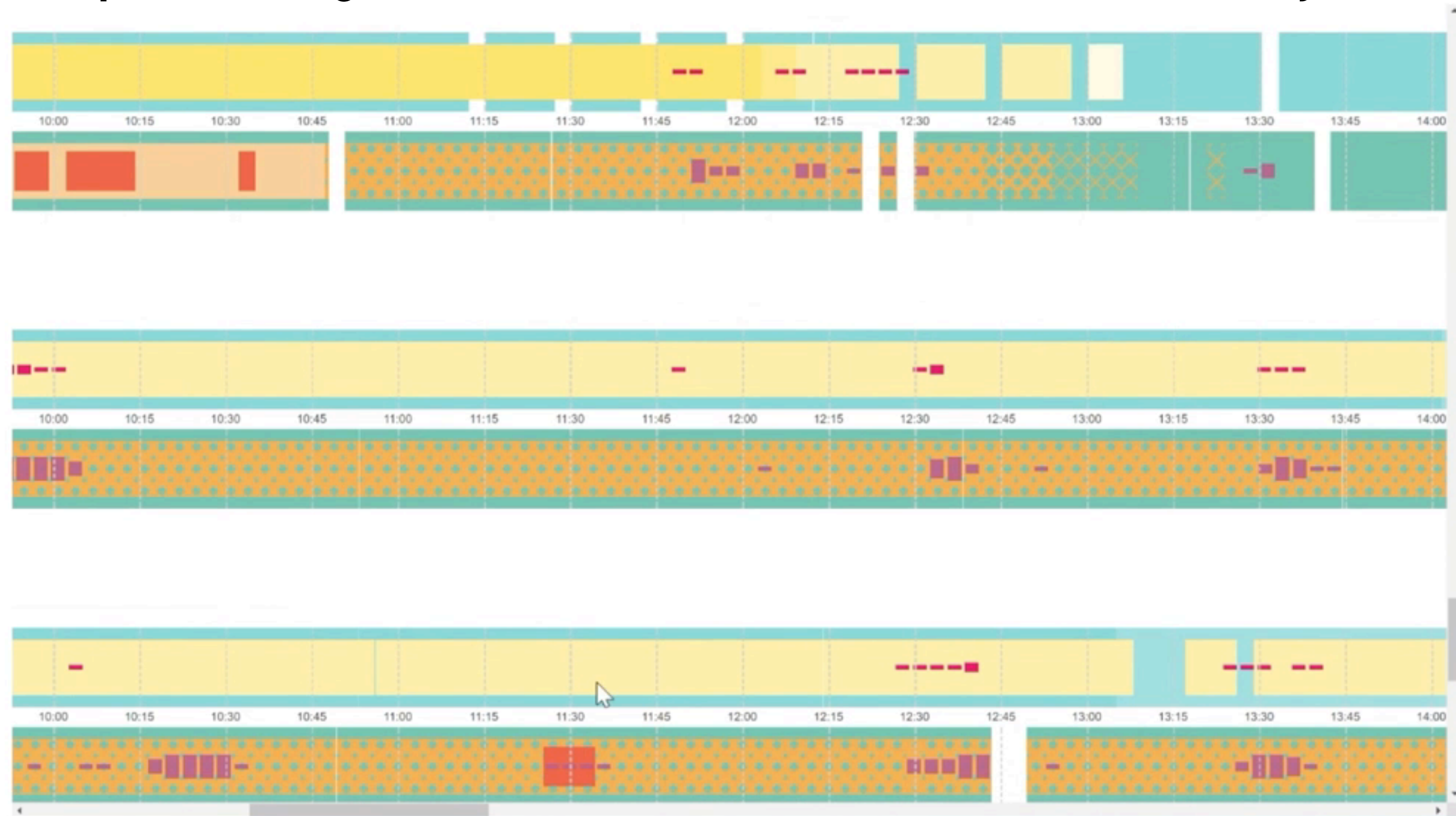




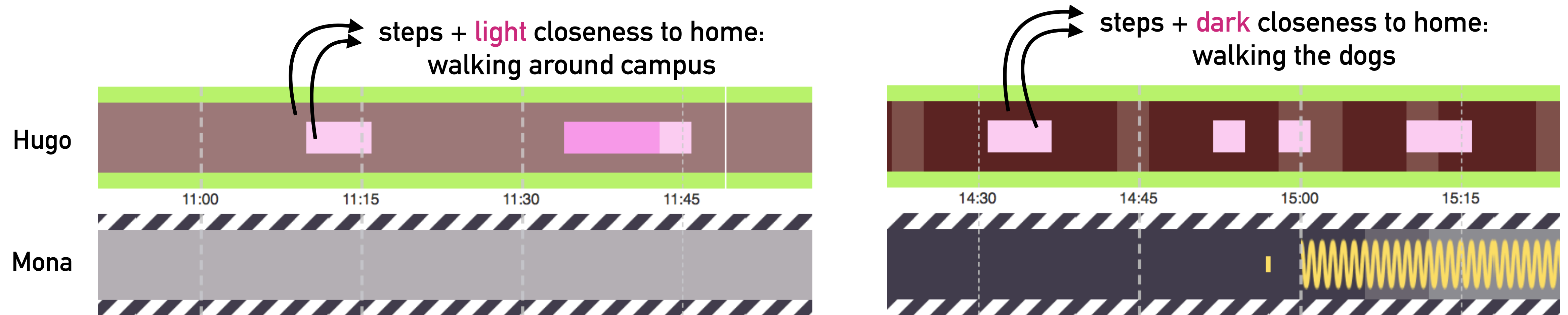


We collected 22 hours of video from 34 Skype interviews

Open coding on videos (Chronoviz) + Thematic Analysis



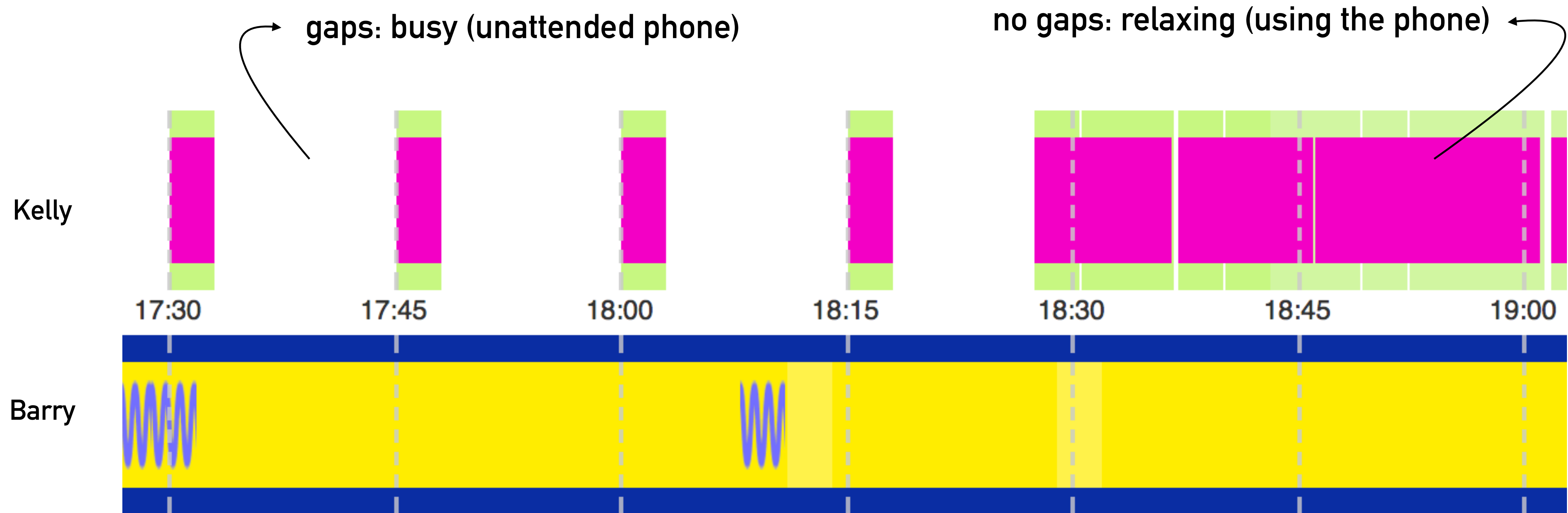
How did **multiple streams** inform context?



Interpreting **multiple streams together** helps disambiguate the meaning of single streams

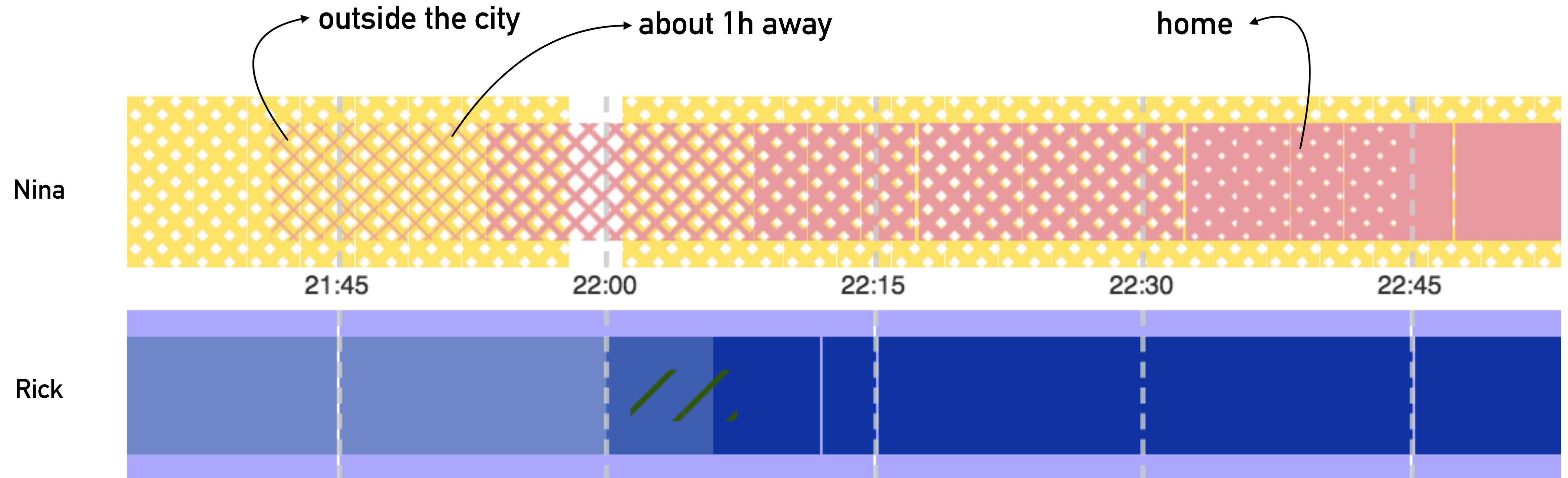
Participants most often looked at **one stream at a time**
according to the data that served their current needs

How did **persistent streams** inform context?



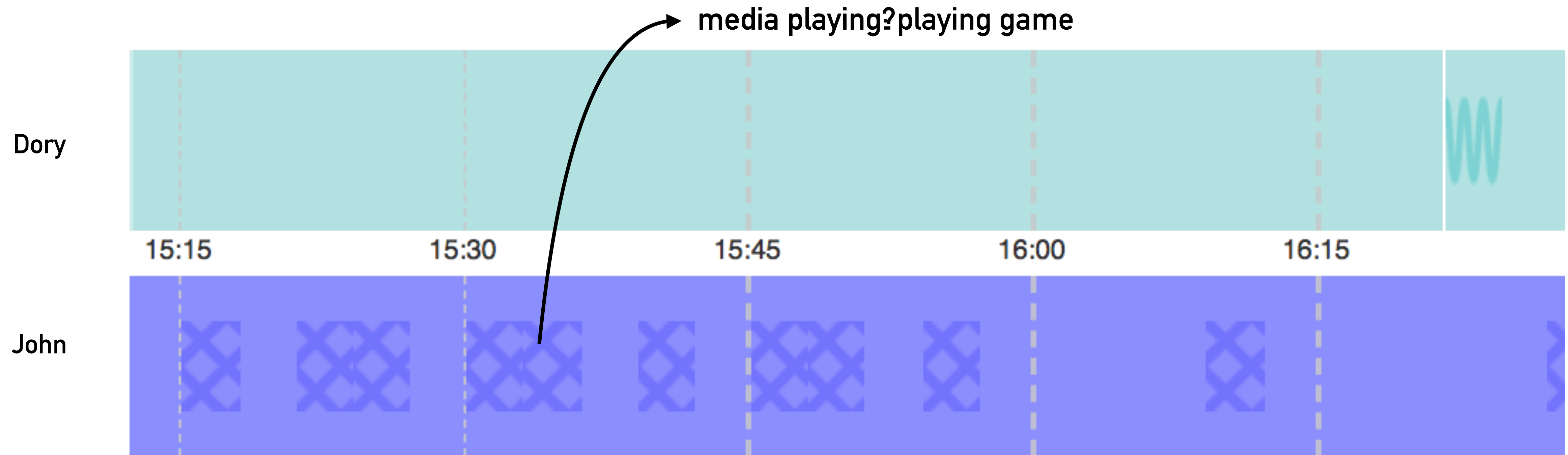
Patterns of missing data provide additional contextual information

How did couples' communication dynamics change?



Contextual information **replaced** direct communication

How did couples' communication dynamics change?



Contextual information **triggered** direct communication

Strong individual differences across and within couples

The streams that triggered or replaced communication depended on the routines, needs and intimate knowledge of each couple

Media Playing

Steps

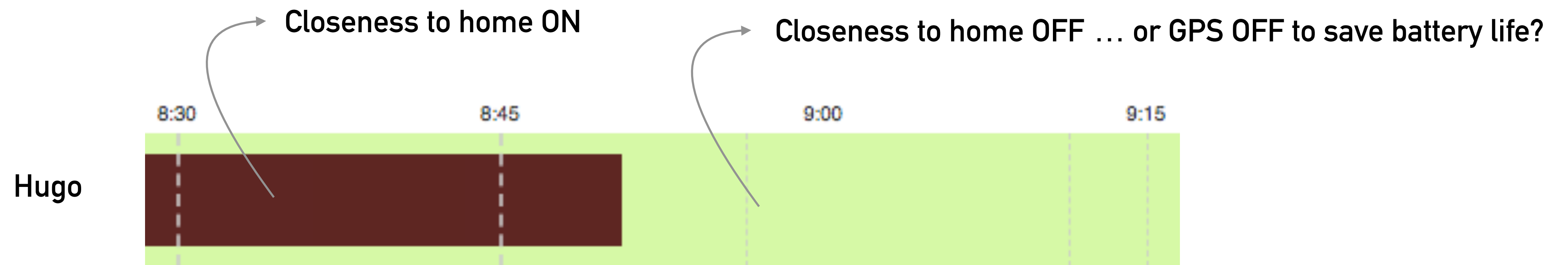
- | | | |
|------------|---|---|
| For Dory: | ask about surprising patterns (trigger) | check that her husband arrived to work safely (replace) |
| For Kelly: | coordinate pick-ups (replace) | warn her husband about his knee health (trigger) |

Data that confirms partners' expectations and knowledge of each other replaces direct communication

Data that challenges partners' expectations and knowledge of each other triggers direct communication

RESULTS

Replacing direct communication
may hurt established communication patterns



Ambiguous visualizations enable **plausible deniability**

A **poor balance between triggering and replacing**
direct communication can lead to feeling more distant

RESULTS

Individual differences around privacy concerns

closeness to home
“I’d like to show him my
exact location to feel safer”
(Mona)

closeness to home
“If I leave work early,
Lifelines spoils the surprise”
(Eva)

too vague

too revealing

data stream

“She’s using Lifelines too
much and stopped texting me”
(Hugo)

“He doesn’t look at it, he doesn’t
care, so it’s not worth it”
(Lucy)

too much

too little

partner's attention
to Lifelines

Final thoughts:

**Why did we choose to do a field study
(technology probe) instead of a usability study?**